

JOSEF SIVIC

INRIA – Willow project
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CURRENT POSITION

Research Scientist (Chargé de recherche), CR1 March 2008-present
Institut National de Recherche en Informatique et en Automatique (INRIA)
WILLOW project, Laboratoire d'Informatique, Ecole Normale Supérieure

EDUCATION

University of Oxford, UK 2002-2006
Ph.D. in Engineering Science
Thesis: *Efficient Visual Search of Images and Videos*
Supervisor: Professor A. Zisserman
Awarded the 2007 BMVA Sullivan Thesis Prize and short-listed (top 3) for the 2007 BCS Distinguished Dissertation award

Czech Technical University, Prague
M.Sc. in Electrical Engineering (with distinction, graduated first in a class of 89)
M.Sc. Thesis: *Geometry of Concentric Multi-perspective Panoramas*

RESEARCH INTERESTS

Visual search, object/person/activity/place recognition, and dynamic 3D scene understanding applied to large image and video collections.

ACADEMIC AND RESEARCH POSITIONS

Postdoctoral Research Associate 2007-2008 (6 months)
CSAIL, Massachusetts Institute of Technology, Cambridge, USA
Research Fellow, University of Oxford, UK Jul. 2006- Sep. 2007
Internship at Microsoft Research, Redmond, USA 3 months, 2005
Student Research Assistant (during MSc), 1997-2002
Center for Machine Perception, Czech Technical University, Prague

TEACHING

Ecole Normale Supérieure, Paris / Ecole Normale Supérieure de Cachan (2009-2014)

- Co-lecturing Object recognition and Computer Vision (M2) and Introduction to computer vision (M1)
- Preparing and grading practical assignments and final class projects.

Co-organizer of a series of international summer schools on Visual Recognition and Machine Learning, which attracted 140-170 participants each year (2010-2013).

<http://www.di.ens.fr/willow/events/cvml2013/>

Lectures at summer schools

- INRIA/ENS Summer school on visual recognition and machine learning, Paris 2013

- INRIA summer school on visual recognition and machine learning, Grenoble 2012
- INRIA/ENS Summer school on visual recognition and machine learning, Paris 2011
- Winter school on computer vision (ENS Lyon, January 2011)
- International computer vision summer school (Sicily, July 2011)
- INRIA summer school on visual recognition and machine learning, Grenoble 2010

University of Oxford, UK

Demonstrating Information Engineering labs for third year undergraduate students:
Image Processing and Signal Analysis Laboratory
Computational Geometry and Estimation Laboratory

SELECTED HONORS AND AWARDS

- **ERC Starting Grant, 2013**
- **Google Research Award, 2012**
- **Prime d'excellence scientifique, INRIA research excellence award, 2009-2016**
- Best poster prize, BMVC'2008
- **Sullivan Thesis Prize, Best Computer Vision PhD thesis in UK, British Machine Vision Association, 2007**
- **Short-listed (top 3) for the Distinguished Dissertation Award, Best Computer Science PhD thesis in UK, British Computer Society, 2007**
- Outstanding reviewer award, BMVC 2008, CVPR 2007
- Best industrial paper prize, British Machine Vision Conference, 2006
- Dean's award for academic achievement, Czech Tech. University, 2002
- Dean's List and National Collegiate Engineering Award, Fall semester 1998

PROFESSIONAL ACTIVITIES

Associate editor, International Journal of Computer Vision (IJCV). IJCV is one of the top two journals in the field.

Area chair for major conferences: CVPR'11, ICCV'11, ECCV'12, CVPR'13, ICCV'13

Invited co-editor of IJCV special issue on "Video Search", 2014.

Peer Reviewer for major conferences and journals in computer vision (ECCV, ICCV, CVPR, PAMI, IJCV), machine learning (NIPS, ICML), computer graphics (SIGGRAPH) and robotics (ICRA, IROS).

Phd Thesis Examiner: M. Johnson, University of Cambridge; M. Bendris, Telecom ParisTech, B. Kaneva, Massachusetts Institute of Technology; N. Ballas, Mines ParisTech

Workshop co-organizer:

MSR-INRIA Workshop on Computer Vision and Machine Learning, January 2010
Workshop on Action Recognition and Pose Estimation in Still Images, ECCV 2012

STUDENTS AND COLLABORATORS

Since joining INRIA in 2008 I have supervised 7 Phd students (1 graduated, 2 more graduating in 2014), 11 MSc students and 5 post-docs.

Post-docs

Bryan Russell, post-doc INRIA, Category-level scene matching and recognition, 2008-2010
Neva Cherniavsky, post-doc MSR-INRIA, Visual attributes for person recognition, 2009-10
Mikel Rodriguez, post-doc DGA/ENS, Visual analysis of crowds in video, 2010-2011
Karteek Alahari, post-doc INRIA, Object and scene recognition in video, 2010- 2013
Visesh Chari, post-doc INRIA, Visual analysis of crowds in video, 2012-

Phd students (with percentage of supervision)

Oliver Whyte, graduated 2012, Removing camera shake blur and unwanted occluders from photographs, 50%

Petr Gronat, 3rd year, Visual place recognition, 50%

Vincent Delaitre, 3rd year, Learning person-object interactions, 50%

Guillaume Seguin, 3rd year, Human pose and action recognition in stereoscopic videos, 50%

Mathieu Aubry, 3rd year, Visual recognition and retrieval of 3D objects and scenes, 80%

Piotr Bojanowski, 2nd year, Learning to annotate dynamic video scenes, 30%

Maxime Oquab, 1st year, Learning to annotate dynamic scenes with convolutional neural networks, 30%

FUNDING I.D.

- ERC Starting Grant LEAP, PI. Funding for 2014-2018: 1.5ME.
- ANR (Agence Nationale de la Recherche) “SEMAPOLIS: Semantic visual analysis and 3D reconstruction of urban environments”, PI. Funding during 2013-2016: 136kE.
- MSR-INRIA project “Understanding Video: understanding, searching, and organizing dynamic video content”. Co-PI. Funding during 2013-2016: 350kE.
- IARPA Finder program, PI. Funding for 2012–2015: 350kE.
- Google Research Award. Co-PI. “Learning to annotate videos from movie scripts”, 2012, 42kE.
- EIT-ICT Labs funded activity “Cross-linking Visual Information and Internet Resources using Mobile Networks”. Co-PI, Funding for 2011-2013: 122kE.
- DGA funded project CrowdChecker. Funding in 2010– 2011: 70kE.
- MSR-INRIA project “Image and Video Mining for Sciences and Humanities”. Funding during 2009-2012: 300kE.

INVITED PRESENTATIONS

34 invited presentations since 2008, including:

Invitations to international workshops: International Workshop on Object Recognition, Lake Como, Italy, 2008; International Workshop on Video, Barcelona, 2009; BIRS Workshop on Computer Vision and the Internet, Banff, Canada, 2010; International Workshop on Search Computing, Brussels, 2012

Invited speaker: Scene Understanding Symposium, MIT, USA, 2008; One day colloquium, Czech Technical University, Prague, 2008 and 2011; AViRS workshop on video surveillance, Paris, 2011; Workshop on Visual Analysis and Geo-Localization of Large-Scale Imagery, ECCV 2012; GdR ISIS workshop, Telecom ParisTech, 2012; Visual analysis beyond semantics, CVPR 2013 workshop; Visual Analysis and Geo-Localization of Large-Scale Imagery, CVPR 2013 workshop; Large-scale video search and mining, ICCV13 workshop; University of Central Florida (2014)

Seminars: University of Oxford (2008); Harvard University (2008) ; Massachusetts Institute of Technology (2008) ; University of Washington (2008) ; Microsoft Research (2008, 2010, 2011) ; University of California, Berkeley (2008, 2012) ; Czech Technical University (2008) ; École nationale supérieure des télécommunications (2008) ; INRIA Grenoble (2009) ; INRIA Rennes (2009) ; Simon Fraser University (2012) ; Google Los Angeles (2012); Carnegie

Mellon University (2012); École Centrale (2013); University of Oxford (2014), New York University (2014); Inria Sophia-Antipolis (2014); KTH Stockholm (2014), MPI Saabracken (2014)

SELECTED SOFTWARE AND TRANSFER

Developed **web-based demo software**¹ for object and person retrieval from entire feature length movies based on (Sivic and Zisserman, 2003) and (Sivic et al., 2005); about 7,000 lines of code.

Since 2008, I have been **consulting** on several projects involving large-scale visual search and classification of images and videos (EADS, ObjectVideo, ProfilTechnology). Building on the follow-up research, mobile visual search company Plink Art was the first UK start-up bought by Google.

Software and demo for non-uniform deblurring from a single image, results of work of my first Phd student O. Whyte are available online².

INRIA SERVICE AND DISSEMINATION

Member of the INRIA “Commission de développement technologique” (CDT), since 2012.

Participant in the INRIA CityLabs initiative, since 2013

Representing INRIA at Salon européen de la Recherche et de l'Innovation (Paris, 2009), CeBit (2012), “Futur en Seine” (2013).

PUBLICATIONS

I have published 40 papers in international journals and major peer-reviewed conferences. The leading conferences in computer vision (ICCV, ECCV, CVPR), machine learning (NIPS) and computer graphics (SIGGRAPH / TOG) have a low acceptance rate (typically below 25%), and publications in their proceedings are considered more important than journal publications. The top journals of the field are the International Journal of Computer Vision (IJCV) and the IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI). Overall, my publications have **over 10,261 citations** and my **h-number is 33** (both obtained from Google Scholar). My publications include:

- 3 book chapters: Springer, Wiley, Cambridge University Press
- 11 journal papers: IJCV (5), PAMI (1), PIEEE (2), IVC (1), SIGGRAPH/TOG (2)
- 35 peer-reviewed conference papers: ICCV (8), CVPR (12), ECCV (8), NIPS (2), BMVC (5)

Three of my conference papers [ECCV'04, BMVC'08, ECCV'12] were **selected for publication in special issues** of IJCV (2006, 2011 and 2014) dedicated to the best (5-6) papers from these conferences.

Publications are available at: <http://www.di.ens.fr/~josef/publications.html>

Book Chapters

[B1] Sivic, J. and Zisserman, A., *Video Google: Efficient Visual Search of Images and Videos*, in Ponce, J., Hebert, M., Schmid, C. and Zisserman, A. (eds), *Toward Category-Level Object Recognition*, Springer-Verlag LNCS Vol. 4170, 2006

¹ www.robots.ox.ac.uk/~vgg/research/vgoogle and www.robots.ox.ac.uk/~vgg/research/fgoogle

² <http://www.di.ens.fr/willow/research/deblurring>

- [B2] Rodriguez, M., Sivic, J., Laptev, I., *Analysis of crowded scenes in video*, Intelligent Video Surveillance Systems, Wiley, 2012
- [B3] Whyte, O., Sivic, J., Zisserman, A., Ponce, J., *Efficient, Blind, Spatially-Variant Deblurring for Shaken Images*, Cambridge University Press, 2014

Journal articles

- [J1] Sivic, J., Schaffalitzky, F. and Zisserman, A., *Object Level Grouping for Video Shots*, International Journal of Computer Vision, 2006
- [J2] Everingham, M. and Sivic, J. and Zisserman, A., *Taking the bite out of automated naming of characters in TV video*, Image and Vision Computing, 2009
- [J3] Sivic, J. and Zisserman A., *Efficient Visual Search Cast as Text Retrieval*, IEEE Transactions on Pattern Analysis and Machine Intelligence, accepted, 2008
- [J4] Sivic, J. and Zisserman A., *Efficient Visual Search for Objects in Videos*, Proceedings of the IEEE, invited paper to the special issue on Advances in Multimedia Information Retrieval, 2008
- [J5] Kaneva, B., Sivic, J., Torralba, A., Avidan, S. and Freeman, W.T., *Infinite Images: Creating and Exploring a Large Photorealistic Virtual Space*, Proceedings of the IEEE, accepted, 2009
- [J6] Philbin, J., Sivic, J. and Zisserman, A., *Geometric Latent Dirichlet Allocation on a Matching Graph for Large-Scale Image Datasets*, International Journal of Computer Vision, 2010
- [J7] Whyte, O., Sivic, J., Zisserman, A. and Ponce, J., *Non-uniform de-blurring for shaken images*, International Journal of Computer Vision, 2012
- [J8] Doersch, C., Singh, S., Gupta, A., Sivic, J. and Efros, A., *What Makes Paris Look like Paris?*, ACM Transactions on Graphics (SIGGRAPH), 31(4), 2012.
- [J9] Aubry, M., Russell, B. and Sivic, J., *Painting-to-3D model alignment via discriminative visual elements*, Transactions on Graphics (TOG), 2014.
- [J10] Whyte, O., Sivic, J., and Zisserman, A., *Efficient, Blind, Spatially-Variant Deblurring for Shaken Images*, International Journal of Computer Vision, 2014.
- [J11] Fouhey, D., Delaitre, V., Gupta, A., Efros, A., Laptev, I. and Sivic, J., *People watching: human actions as a cue for single-view geometry*, International Journal of Computer Vision, 2014.

Reviewed Conference Proceedings

- [C1] Sivic, J. and Zisserman, A., *Video Google: A Text Retrieval Approach to Object Matching in Videos*, International Conference on Computer Vision (ICCV), Oct 2003, **(Over 2500 citations on Google Scholar)**
Demo: <http://www.robots.ox.ac.uk/~vgg/research/vgoogle>
- [C2] Sivic, J., Schaffalitzky, F. and Zisserman, A., *Object Level Grouping for Video Shots*, European Conference on Computer Vision (ECCV), May 2004, **Oral presentation**
- [C3] Sivic, J. and Zisserman, A., *Video Data Mining Using Configurations of Viewpoint Invariant Regions*, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), June 2004, **Oral presentation**
- [C4] Sivic, J., Everingham, M. and Zisserman, A., *Person spotting: video shot retrieval for face sets*, International Conference on Image and Video Retrieval (CIVR), 2005, **Oral presentation**
Demo: <http://www.robots.ox.ac.uk/~vgg/research/fgoogle>

- [C5] Sivic, J., Russell, B. C., Efros, A. A., Zisserman, A. and Freeman, W. T., *Discovering Object Categories in Image Collections*, Proceedings of the International Conference on Computer Vision (ICCV), 2005, Together with associated technical report: **over 1000 citations on Google Scholar**
- [C6] Russell, B. C., Efros, A. A., Sivic, J., Freeman, W. T. and Zisserman, A., *Using Multiple Segmentations to Discover Objects and their Extent in Image Collections*, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2006
- [C7] Sivic, J., Zitnick, C. L. and Szeliski, R., *Finding people in repeated shots of the same scene*, Proceedings of the British Machine Vision Conference (BMVC), 2006, **Oral presentation**
- [C8] Everingham, M., Sivic, J. and Zisserman, A., *Hello! My name is... Buffy -- Automatic Naming of Characters in TV Video*, Proceedings of the British Machine Vision Conference (BMVC), 2006, **Oral presentation**
- [C9] Philbin, J., Chum, O., Isard, M., Sivic, J. and Zisserman, A., *Object retrieval with large vocabularies and fast spatial matching*, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2007, **Over 800 citations on Google Scholar**
- [C10] Chum, O., Philbin, J., Sivic, J., Isard, M. and Zisserman, A., *Total Recall: Automatic Query Expansion with a Generative Feature Model for Object Retrieval*, Proceedings of the International Conference on Computer Vision (ICCV), 2007
- [C11] Sivic, J., Russell, B. C., Zisserman, A., Freeman, W. T. and Efros, A. A., *Unsupervised Discovery of Visual Object Class Hierarchies*, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2008
- [C12] Philbin, J., Chum, O., Isard, M., Sivic, J. and Zisserman, A., *Lost in Quantization: Improving Particular Object Retrieval in Large Scale Image Databases*, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2008
- [C13] Philbin, J., Sivic, J. and Zisserman, A., *Geometric LDA: A Generative Model for Particular Object Discovery*, Proceedings of the British Machine Vision Conference (BMVC), 2008
- [C14] Liu, C., Yuen, J., Torralba, A., Sivic, J. and Freeman, W. T., *SIFT Flow: Dense Correspondence across Different Scenes*, Proceedings of the European Conference on Computer Vision (ECCV), 2008, **Oral presentation**
- [C15] Sivic, J., Everingham, M. and Zisserman, A., *"Who are you?": Learning person specific classifiers from video*, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2009
- [C16] Whyte, O., Sivic, J. and Zisserman, A., *Get Out of my Picture! Internet-based Inpainting*, Proceedings of the British Machine Vision Conference (BMVC), 2009, **Oral presentation**
- [C17] Duchenne, O., Laptev, I., Sivic, J., Bach, F. and Ponce, J., *Automatic Annotation of Human Actions in Video*, Proceedings of the International Conference on Computer Vision (ICCV), 2009
- [C18] Russell, B., Efros, A., Sivic, J., Freeman, W. T. and Zisserman, A., *Segmenting Scenes by Matching Image Composites*, Advances in Neural Information Processing Systems (NIPS), 2009
- [C19] Whyte, O., Sivic, J., Zisserman, A., Ponce, J., *Non-uniform de-blurring for shaken images*, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2010, **Oral presentation**
- [C20] Delaitre, V., Laptev, I., Sivic, J., *Recognizing human actions in still images: a study of bag-of-features and part-based representations*, British Machine Vision Conference (BMVC), 2010

- [C21] Knopp, J., Sivic, J. Pajdla, T., *Avoiding confusing features in place recognition*, European Conference on Computer Vision (ECCV), 2010
- [C22] Philbin, J., Isard, M., Sivic, J., Zisserman, A., *Descriptor learning for efficient retrieval*, European Conference on Computer Vision (ECCV), 2010, **Oral presentation**
- [C23] Lezama, J., Alahari, K., Sivic, J. and Laptev, I. *Track to the future: Spatio-temporal video segmentation with long-range motion cues*, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2011
- [C24] Rodriguez, M., Laptev, I., Sivic, J. and Audibert, J.-Y. *Density-aware person detection and tracking in crowds*, IEEE International Conference on Computer Vision (ICCV), 2011
- [C25] Rodriguez, M., Sivic, J., Laptev, I. and Audibert, J.-Y., *Data-driven crowd analysis in videos*, IEEE International Conference on Computer Vision (ICCV), 2011, **Oral presentation**
- [C26] Delaitre, V., Sivic, J. and Laptev, I., *Learning person-object interactions for action recognition in still images*, Advances in Neural Information Processing Systems (NIPS), 2011
- [C27] Delaitre, V., Fouhey, D., Efros, A., Gupta, A., Laptev, I. and Sivic, J., *People Watching: Human Actions as a Cue for Single View Geometry*, European Conference on Computer Vision, 2012, **Oral presentation**
- [C28] Delaitre, V., Fouhey, D., Laptev, I., Sivic, J., Efros, A., Gupta, A., *Scene semantics from long-term observation of people*, European Conference on Computer Vision, 2012.
- [C29] Gronat, P., Obozinski, G., Sivic, J. and Pajdla, T., *Learning per-location classifiers for visual place recognition*, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2013
- [C30] Torii, A., Sivic, J., Pajdla, T. and Okutomi, M., *Visual place recognition with repetitive structures*, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2013
- [C31] Bojanowski, P., Laptev, I., Ponce, J., Schmid, C. and Sivic, J., *Finding actors and actions in movies*, IEEE International Conference on Computer Vision (ICCV), 2013
- [C32] Alahari, K., Seguin, G., Laptev, I. and Sivic, J., *Pose estimation and segmentation of people in 3D movies*, IEEE International Conference on Computer Vision (ICCV), 2013
- [C33] Aubry, M., Efros, A., Russell, B. and Sivic, J., *Seeing 3D chairs: exemplar part-based 2D-3D alignment using a large dataset of CAD models*, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2014
- [C34] Oquab, M., Bottou, L., Laptev, I. and Sivic, J., *Learning and transferring mid-level image representations using convolutional neural networks*, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2014
- [C35] Bojanowski, P., Lajugie, R., Bach, F., Laptev, I., Ponce, J., Schmid, C. and Sivic, J., *Weakly Supervised Action Labeling in Videos Under Ordering Constraints*, European Conference on Computer Vision (ECCV), 2014
- [C36] Vu, T.-H., Olsson, C., Laptev, I., Oliva, A. and Sivic, J. *Predicting Actions from Static Scenes* European Conference on Computer Vision (ECCV), 2014

Pre-prints

- [P1] Oquab, M., Bottou, L., Laptev, I. and Sivic, J., *Weakly Supervised Object Recognition with Convolutional Neural Networks*, <http://www.di.ens.fr/willow/research/weakcnn/>

Reviewed workshops

- [W1] Sivic, J., Kaneva, B., Torralba, A., Avidan, S. and Freeman, W.T., *Creating and Exploring a Large Photorealistic Virtual Space*, First IEEE Workshop on Internet Vision (in conjunction with CVPR), 2008
- [W2] Cherniavsky, N., Laptev, I., Sivic, J., Zisserman, A., *Semi-supervised learning of facial attributes in video*, First International Workshop on Parts and Attributes, European Conference on Computer Vision (ECCV), 2010, **Oral presentation**
- [W3] Kaneva, B., Sivic, J., Torralba, A., Avidan, S. and Freeman, W. T., *Matching and predicting street level images*, Workshop on Vision for Cognitive Tasks, European Conference on Computer Vision, 2010
- [W4] Russell, B., Sivic, J., Ponce, J. and Dessales, H., *Automatic alignment of paintings and photographs depicting a 3D scene*, 3rd International IEEE Workshop on 3D Representation for Recognition, 2011
- [W5] Torii, A., Sivic, J. and Pajdla, T., *Visual localization by linear combination of image descriptors*, 2nd IEEE Workshop on Mobile Vision, 2011
- [W6] Whyte, O., Sivic, J. and Zisserman, A., *Deblurring shaken and partially saturated images*, IEEE Workshop on Color and Photometry in Computer Vision, 2011

Invited Papers

- [I1] Sivic, J., Schaffalitzky, F. and Zisserman, A., *Efficient Object Retrieval from Videos*, European Signal Processing Conference (EUSIPCO), Vienna, Austria, 2004, (invited to special session on affine covariant regions), **Oral presentation**
- [I2] Sivic, J. and Zisserman, A., *Efficient Visual Content Retrieval and Mining in Videos*, Pacific-rim conference on Multimedia (PCM), Tokyo, Japan, 2004, **Oral presentation**

Technical Reports

- [T1] Sivic, J. and Pajdla, T., *Geometry of Concentric Multiperspective Panoramas*, M.Sc. thesis and technical report, Czech Technical University, February 2002.
- [T2] Philbin, J. , Bosch, A. , Chum, O. , Geusebroek, J. , Sivic, J. and Zisserman, A., *Oxford TRECVID 2006 - Notebook Paper*, November 2006